Final exam:

5 Questions:

- Q1. 45 → 11
- Q2 15
- Q3 15
- Q4 15
- Q5 10

Numerical integration:

(N-C). Trapezoidal rule. Simpson's rule. Midpoint rule

+ error terms + analysis of the error

+ Composite rules (error)

Gaussian quadrature. (derivation ?

$$\int_a^b f(x) = \sum_{i=1}^{n} w_i f(x_i)$$

degree of precision , orthogonal polynomials

Richardson Extrapolation ? (definition, algorithm)

Numerical linear algebra.

nonsingular / pivoting / partial pivoting / total ...

permutation matrix / orthogonal matrix / p. d. /

norm (induced norm) (Fro... $\|A\|_2$ $\|A\|_1$ $\|A\|_\infty$)

condition number / convergent matrix /.

Gaussian Elimination $\Longleftrightarrow$ LU Factorization

How to compute?

How to use?

LL ... (P. d.)

↓

Corut ... (tridiagonal)

(Forward substitution. backward ...)

Iterative method:

(Jacobi, Gauss-Seidel, SOR)

$x^{(k+1)} = Tx^{(k)} + c$

+ convergence ($\rho(A) < 1$) + error estimate $\|x^{(k)} - x\| \le \frac{\|T\|^k}{1-\|T\|}(x^{(1)} - x^{(0)})$

$\rho(A) \le \|A\|$ for any induced norm $\Leftarrow$?

definition

Proofs related to the condition number.